Question ID: 4b3d6062

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

The *Mammillaria* cactus *M. boolii* occurs naturally only in the state of Sonora in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. boolii*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of *M. boolii* could thus ______

Which choice most logically completes the text?

Answer

A. lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.

B. allow conservationists to better consider how climatic changes affecting Sonora may reduce the number of species competing with *M. boolii*.

C. erroneously shift the focus of conservation efforts away from *M. boolii* itself.

D. fail to consider the ways in which the survival of *M. boolii* may be influenced by changes in the populations of other species that inhabit Sonora.

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text's discussion of conservation approaches for the *Mammillaria* cactus *M. boolii*. The text establishes that *M. boolii* only grows naturally in the state of Sonora in Mexico, which makes it particularly vulnerable to extinction. The text then contrasts two approaches to conservation: the traditional single-species approach that individually focuses on at-risk species and a newer ecosystem-based approach that considers the interactions between climate, terrain, and various species in a geographical area. According to the text, conservationists have recently argued that this ecosystem-based approach may lead to better outcomes for all species in a location. If this ecosystem view is correct, then the single-species approach to conserving *M. boolii* would likely fail to consider how the cactus's survival depends on its interactions with other species in Sonora's ecosystem.

Choice A is incorrect because the text doesn't address the distribution of *Mammillaria* species besides *M. boolii* throughout Mexico or discuss how that distribution affects survival. The text focuses specifically on *M. boolii* in Sonora and different approaches to its conservation. Choice B is incorrect because the text doesn't suggest that climatic changes in Sonora would reduce competition for *M. boolii* or that conservation efforts are focused on understanding this specific dynamic. In fact, the text implies that an ecosystem-based approach would consider climate among many other factors but doesn't specify how climatic changes might affect competition between species. Choice C is incorrect because the text doesn't suggest that the single-species approach would shift the focus of conservation efforts away from *M. boolii* itself—rather, it suggests that this approach might be too narrowly focused on *M. boolii* alone without considering the broader ecosystem-related factors that affect its survival.